

PAPER_858: Westerlund 2 Quadriadic UQFF Four-Set Simultaneous Real and Imaginary Solutions

Author: Daniel T. Murphy - Star Magic / UQFF Framework **Date:** 2026-04-04 **Session:** 199 **Source:** grok_share_589f6949-6fe9.txt (2404 lines, May 05 - Aug 07, 2025) **Calculator:** Westerlund2QuadriadicRealImaginaryCalc (CP4 #442) **CVW:** v2.0.0 compliant

Abstract

We solve the full quadriadic UQFF system (four master equations) simultaneously for Westerlund 2, a massive star cluster in Carina ($\sim 20,000$ ly, age ~ 2 Myr). The four equations - (1) Compressed Gravity F_{U_g} , (2) Resonance $R(t)$, (3) Buoyancy $F_{U_{Bi}}$, (4) Universal Magnetism U_m - each yield both real and imaginary solutions. The imaginary components represent the quantum superconducting portion of each field. For Westerlund 2: $F_{U_g} \sim 2.43e-40$ N, $R(t) \sim 6.67e-41$ N, $F_{U_{Bi}} \sim 6.14e-32$ N, $U_m \sim 1.80e5$ J/m³. Buoyancy dominates the force balance, consistent with the region's intense star formation activity.

1. Core Equations

- $F_{U_g} = \sum_k [k_k (f_{UA} * f_{SCm} * REB)^2 / r^2 * G_k] + k_4 \rho_{vac} M_{BH} / r * \exp(-\alpha * t) * \cos(\pi * t / n) * (1 + f_{feedback})$
 - $R(t) = \sum_{i=1}^{26} [R_{Ug1,i} \cos(w_1 * t) + R_{Ug2,i} \cos(w_2 * t) + R_{Ug3,i} \cos(w_3 * t) + R_{Ug4,i} \cos(w_4 * t)]$
 - $F_{U_{Bi}} = \sum_k [k_{Ub,k} * f_{UA} * f_{SCm} * REB / r^2 * H_k * f_{Ub}]$
 - $U_m = \sum_j [\mu_j(t) / r_j * (1 - \exp(-\gamma * t) * \cos(\pi * t / n)) * \phi^j] * P_{SCm} * E_{react} * (1 + 1e13 * f_H) * (1 + f_q)$
 - $f_{Ub} = k_{Ub} * \Delta k_{eta} * (\rho_{vac}, [UA] / \rho_{vac}, [SCm]) * (V_{little} / V_{big})$
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2. UQFF Integration

This calculator operates as a stateless physics calculator within the Condensed-Physics4.py (Phase 4) IPC chain. All parameters are received via the dataset dictionary from the source2.cpp principal GUI pipeline. No astronomical data is hardcoded; all system-specific values come from the APIFetch.py -> bodies_*.csv data flow.

3. Source Data

- File:** grok_share_589f6949-6fe9.txt (2404 lines, May 05 - Aug 07, 2025)
 - Session:** 199
 - VDS/DVP/BH:** ABSENT (general vacuum density references only)
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References

- Murphy, D.T. - Star Magic UQFF Framework (2024-2026)
- Srivastava, Y.N., Widom, A., Larsen, L. - Electroweak neutron production (LENR)

3. Kepler Mission DR25 - 4,034 candidates, 2,335 confirmed planets
4. Hubble Heritage Team / A. Nota (ESA/STScI) - Westerlund 2 / NGC 346 imaging
5. UQFF Calibration: $\kappa=0.0005/\text{day}$, $[\text{SSq}]=0.57$, $\beta_i \sim 0.603$